

Binary · IG · Answers

24 written + 6 pen-first = 30 · Student copy

One mark per stated point unless grouped. Accept equivalent correct working. Do not credit contradictory working; TV = too vague, NE = not enough. Width marks apply only where stated below.

1. Counter [3]

(a) 0110, 0111, 1000, 1001. First two correct [1]; last two correct [1]. (b) 8 [1].

2. Denary to binary [4]

(a) $53 = 32 + 16 + 4 + 1$ [1] → **00110101** [1]. (b) $170 = 128 + 32 + 8 + 2$ [1] → **10101010** [1]. Accept a correct repeated-division method for each working point.

3. Switch [1]

Binary has two digits/states corresponding to the switch's on/off states [1]. "Computers use binary" alone is NE.

4. Binary to denary [4]

(a) $64 + 16 + 4 + 1$ [1] = **85** [1]. (b) $128 + 64 + 32 + 8 + 2 + 1$ [1] = **235** [1].

5. Pool [2]

0000000100101100. Correct binary value ($256 + 32 + 8 + 4 = 300$) [1]; correct value padded to exactly 16 bits [1].

6. Capacity [2]

(a) **256** or 2^8 [1]. (b) **255** [1].

7. Authored visitor-counter rehearsal [4]

(a) $73 \rightarrow$ **01001001** [1]; $198 \rightarrow$ **11000110** [1]. (b) **255** [1]. (c) **12** [1]: eleven bits reach 2047; twelve reach 4095. Fresh values; authored allocation. Explanations here are guidance, not extra required mark points.

8. Addition and overflow [4]

(a) Correct carry working ($200 + 100 = 300$) [1]; full sum **100101100** [1]. (b) An unsigned eight-bit register holds at most **255** [1]; 300 exceeds this range / requires nine bits [1]. Overflow is the failure of the true result to fit; it is not defined as always wrapping silently.

9. Pen-first card task [6]

(a) YES, YES, NO, YES, YES, NO; $27 = 16 + 8 + 2 + 1$. Correct sum of weights [1]; all six YES/NO answers correct [1]. (b) **101010** [1]; $32 + 8 + 2 =$ **42** [1]. (c) New weight **64** [1]; maximum **127** [1]. Zero is the all-NO pattern.

Extension (unmarked): $022_3 = 2 \times 3 + 2 = 8$; the next count is $100_3 = 9$. Each position carries when it reaches three because its allowed digits are only 0, 1, 2. Karel companion: outside the mark total.